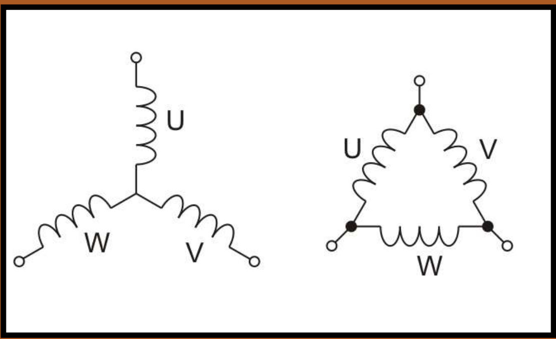
Worksheet 18

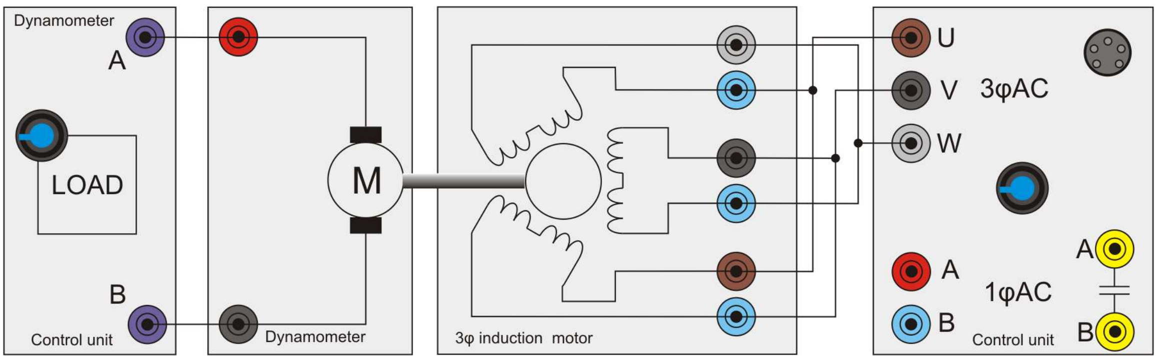
Delta configuration

Modern electrical machines



Induction motors can be used in 'star' or 'delta' configuration. This terminology just refers to the way the windings are connected to the supply. Delta connection tends to be used when higher current or higher power is needed.

Image shows the wiring connections of star and delta.



Over to you:

1) Set up the three phase induction motor as shown in the diagram above.
2) On your PC, load 'Open Log_3phase.bat'.
3) Click on the '20V/12V' switch. In Delta configuration the resistance of the windings to the power supply is much lower and the control box will trip if you draw too much current.
4) With an output frequency of 50Hz, run the program and take an automatic set of readings over a range of dynamometer loads.
4) Repeat this for output frequencies of 60, 70, 80, 90, 100 Hz.
5) If you can plot all output lines on the same graph.
6) On the next page, sketch the results .

How do you change direction of rotation?

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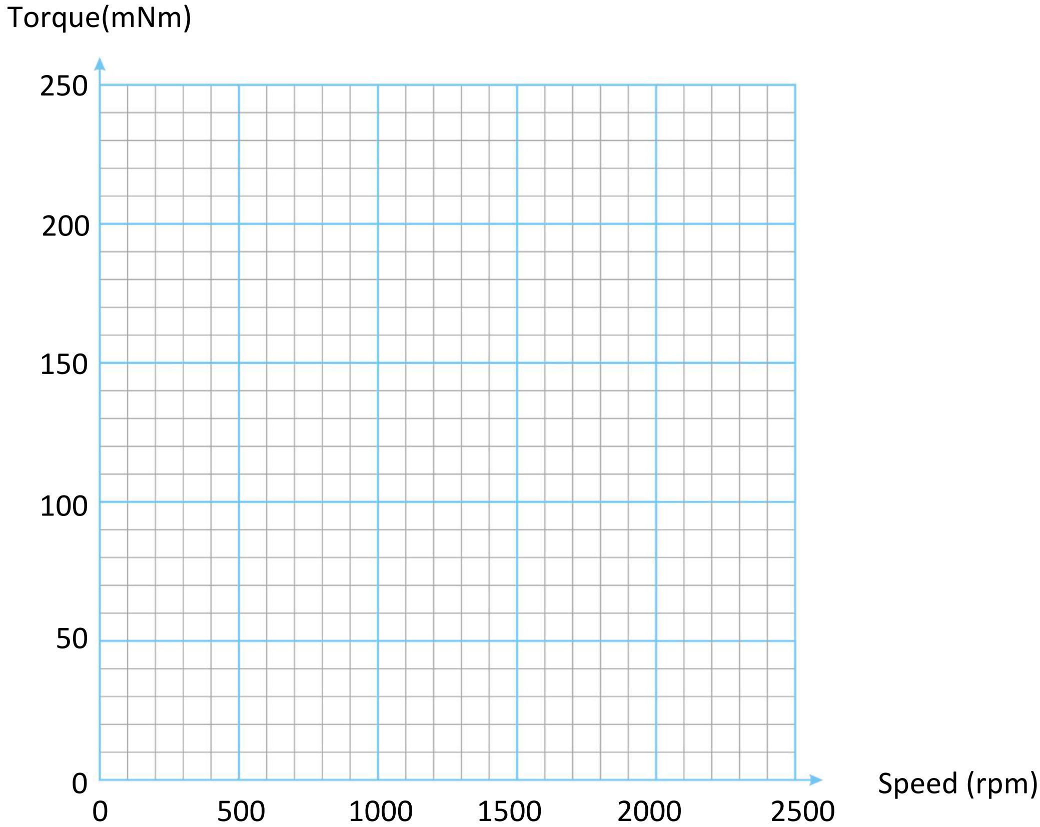

The effect of speed on torque for different values of output frequency.

So what?

What can you say about the power consumed and the torque of both star and delta configurations.